

Like terms



1 State whether the following pairs are like terms:

a $9y$ and $10y$

b $9n^2$ and $2n$

c $2xy$ and $4yx$

d $10x^2y$ and $9y^2x$

e $6x^3y$ and $10yx^3$

f $13uv$ and $17vu$

g $10z$ and $10b$

h $8y$ and $5y$

2 Simplify the following by collecting like terms:

a $2a + 5a$

b $10x + 6x$

c $4b + 3b$

d $12y - 3y$

e $3c + 4c + 7c$

f $15x - 6x - 2x$

g $19b - 12b - 6b$

h $-3n + 6n + 3n$

i $8x - 3x + 7x$

j $20x^2 - 12x^2 + 10x^2 - x^2$

k $4xy + 10xy - 9xy$

l $10ab + 3ab - 8ba$

Simplify expressions

3 Consider the expression: $4y - 3y$.

a Simplify the expression.

b Explain why the answer to part (a) was not $1y$.

4 For each of the following expressions:

i State the pairs of like terms.

ii Hence, simplify the expression.

a $5u + 7 + 8u + 8$

b $7u + 2v - 7u$

c $3p + 5p^2 - 3p^2 + p$

d $4x - 6 + 5x$

5 Simplify:

a $8x + 6y - 2y - 4x$

b $10m + 9n + 5m + 10n$

c $10x - 9y - 7x + 11y$

d $11m + 8n + 14m$

e $9xy + 12yx$

f $6p + 8q - 6p$

g $9p^2 - 11p + 11p^2$

h $7n^2 + n^2 - 4m + 2m^2 - 5n^2$

i $8x + 9y - 5x + 10y$

j $7a + 11a - 9b + b$

k $12x^2 + 10x - 9x - 5x^2$

l $8x - 7y - 6z + 10z$

m $15xy - 8x^2 + 10xy - x^2$

n $10a - 8b + 4a - 8a + b$

- o** $2ab - 7c + ab + 9c$
- q** $18x - 9y + 7x + 11y$
- s** $13m + 2n - 8m + 10n$



- p** $12j + 9k - 7k + 11j$
- r** $11y - 6z + y + 6z$
- t** $-3s + 4t - 6t + 9s$

6 Simplify:

- a** $5x + 2y - 8x + 6z - 3y + 4z$
- b** $2xy + 5yz + 4xz - 8xy + 2xz - 3yz$
- c** $8x^2 + 10x - 6 + 4x^2 - x + 11$
- d** $20 - 9x + 4y + 6y - 7x + 21$
- e** $9mn + 8m - 7n + 7n - 10m + 2mn$
- f** $-6 + 8x^2 - 9 + 5x + 16x^2$
- g** $11s + 6t + 3st - 2ts + t - 4s$
- h** $13ab - 8bc + 6ba - 9b + 11cb$

7 Complete the following statements:

- a** $2x + \square = 8x$
- b** $7x - \square = 2x$
- c** $9x + 4y + \square = 9x + 6y$
- d** $7x + \square + 10y - \square = 11x + 6y$
- e** $5x^2 - 2x + \square + \square = 9x^2 + 3x$
- f** $8ab + \square + 9b + 6ab = \square + 11b$